

Banid Prajapati

Kathmandu, NP | banidprajapati98@gmail.com | +977-9847297346

SUMMARY

Applied AI and Machine Learning specialist with hands-on experience in research-driven AI systems, RAG pipelines, OCR workflows, and intelligent automation. Skilled in Python, embeddings, vector databases, data processing, and LLM-based applications, with experience transforming large-scale structured and unstructured data into actionable insights through scalable AI solutions.

EXPERIENCE

Vertex Special Technology | AI/ML Engineer

Kathmandu, NP | 07/2025 – Current

- Researched and engineered HEM-Screen, an AI-based anemia detection system using eye conjunctiva, palm, and nail images. Designed and trained 4 models under lead mentor supervision:
 - Faster R-CNN and Quantized Faster R-CNN : 98% and 94% accuracy
 - UNet: 96% accuracy
 - ResNet-based multi-modal model combining images and demographic metadata: 80% accuracy
- Developed a synthetic Nepali handwriting data generation system by decomposing words into reusable handwritten character primitives, expanding a limited corpus of 48 handwritten tokens into 15,000 unique word samples.
- Conducted document intelligence research and developed OCR-based check detection and feature validation models for financial datasets, achieving 95% real-world accuracy.
- Researched and implemented an embedding-based face recognition attendance system using vector similarity search for real-time identity matching across multiple CCTV feeds.
- Led a 12-member data annotation team, establishing labeling standards and validating OCR training datasets to maintain 97% annotation accuracy for machine learning research workflows.

Rooster Logic | Python Internship

Kathmandu, NP | 05/2024 – 08/2024

- Researched and built Nepali audio to English transcription script using Google Speech-to-Text API increasing the speed from manual work to 80% faster.
- Automated Google Docs template formatting with Apps Script, reducing manual work by 99%.
- Developed Nepali-to-English translation script using Google Cloud Translation API saving total of 200 hours of manual data annotation.

SKILLS

Technical Skills: LLMs | Prompt Engineering | AI Agents | RAG Pipelines | Llamaindex | LangGraph | Hugging Face Transformers | LLM Fine-tuning | Embeddings | Vector Databases | Python | PyTorch | scikit-learn | FastAPI | REST APIs | Docker | Pandas | SQL | MongoDB | Node.js | React | Git

Soft Skills: Problem-solving | Critical thinking | Fast learner | Adaptability | Analytical thinking | Communication | Ownership mindset | Teamwork | Time management | Leadership

EDUCATION

BSc (Hons) Computer Science

Kathmandu, NP | 11/2022 – 06/2025

Herald College Kathmandu | University of Wolverhampton

- Awarded AAA scholarship for outstanding academic, attention and attitude performance.

Data Fellowship 2025 – Code for Nepal x DataCamp Donates

Kathmandu, NP | 04/2025 – 04/2026

- Awarded access to DataCamp's comprehensive data science training program through Code for Nepal's initiative.
- Engaged in self-paced learning covering data analysis, Python programming, machine learning, and related data skills.

NOTABLE INDIVIDUAL PROJECTS

Nepal Constitution RAG System – Legal Document Q&A | Open Source Project

- Researched and engineered a Retrieval-Augmented Generation (RAG) system for querying Nepal's Constitution using FastAPI, LlamaIndex, and Qdrant vector database.
- Engineered semantic chunking strategy with sentence-based splitting, overlap continuity, and hierarchical Part/Article metadata for improved retrieval.
- Integrated Hugging Face's multilingual-e5-base embeddings and Llama-3-8B for accurate legal document retrieval and response generation.
- Built evaluation pipeline using RAGAS metrics achieving ~80% overall accuracy.

Document Segmentation Model | Open Source Project

- Built a deep learning pipeline to detect and extract documents (IDs, licenses, certificates) from real-world images using pixel-level segmentation.
- Trained a U-Net with MobileNetV2 encoder on DocCornerDataset (HuggingFace) for robust document boundary prediction.
- Achieved strong performance: IoU 0.8886 | Dice 0.9393 | Pixel Accuracy 0.9549.
- Designed an end-to-end workflow (train, eval, inference) with modular PyTorch components for reproducible experimentation.
- Converted segmentation masks into OCR-ready document crops via post-processing and perspective correction.

NOTABLE EXPERIENCES

WOW Nepal | Volunteer

Kathmandu, NP | 11/2024

- Conducted a photography session for 28 deaf students.
- Promoted inclusive learning techniques.
- Assisted logistics at large-scale events, including WOW Festival by British Council.

Antarikhya Pratisthan Nepal | Volunteer

Kathmandu, NP | 4/2024

- Supported program operations for Space Day.
- Managed backstage event logistics for smooth execution.

UI Visuals Community | Steering Leader at Herald College Kathmandu

8/2024 – 10/2025

- Led a 26-member creative team.
- Directed concept development, design strategy, and event execution.
- Orchestrated 8 large-scale events at Herald DevFest 2025 with 1,400+ participants.
- Achieved 2nd place overall at DevFest 2025.
- Curated programs for college students, orphanages, and +2 institutions.

DesignXplore 2024 | Event Organizer at Herald College Kathmandu

6/2024 – 8/2024

- Organized intercollege competition on design, sustainability, and creativity.
- Managed 54 participants from 7 colleges.
- Compiled a 50-page report with participant feedback and judge evaluations.

Herald DevFest 2023 | Winner & Organizer at Herald College Kathmandu

8/2023 – 10/2023

- Managed 7 events in 1 month for 800+ participants.
- Drafted detailed proposals for each event.
- Designed promotional materials in Illustrator, increasing engagement and participation.